

HomeWork #5

Design & Analysis of Algorithms

Name: Muhammad Hassan Khalid

ROLL-NO: 21L-5692

Task 1:

Rod Length = 6

Price = 17

Lengths = Price

- ① $5 + 1 = 11$
- ② $4 + 2 = 14$
- ③ $4 + 1 + 1 = 11$
- ④ $3 + 2 + 1 = 14$
- ⑤ $3 + 1 + 1 + 1 = 11$
- ⑥ $3 + 3 = 16$
- ⑦ $2 + 2 + 2 = 15$
- ⑧ $2 + 2 + 1 + 1 = 12$
- ⑨ $2 + 1 + 1 + 1 + 1 = 9$
- ⑩ $1 + 1 + 1 + 1 + 1 + 1 = 6$

The best price can be achieved by not cutting the rod at all it will be of the highest value that is 17.

TASK 2

Recursive Equation:-

$$\text{MaxValue}(\text{Length}) = \max(\text{price}[i] + \text{MaxValue}(\text{Length} - i))$$

- Rod length = 1;

$$\text{MaxValue}(1) = \max(1) = 1$$

- Rod length = 2;

$$\begin{aligned}\text{MaxValue}(2) &= \max(5, \text{MaxValue}[1] \\ &+ \text{MaxValue}[1]) = \max(5, 1+1) \\ &= 5\end{aligned}$$

- Rod length = 3;

$$\begin{aligned}\text{MaxValue}(3) &= \max(8, \text{MaxValue}(1) + \\ &\text{MaxValue}(2), \text{MaxValue}(2) + \\ &\text{MaxValue}(1)) \\ &= \max(8, 1+5, 5+1) \\ &= 8\end{aligned}$$

- Rod length = 4;

$$\begin{aligned} \text{MaxValue}(4) &= \max(9, \text{MaxValue}(1) + \text{MaxValue}(3), \text{MaxValue}(2) + \text{MaxValue}(2), \\ &\quad \text{MaxValue}(3) + \text{MaxValue}(1)) \\ &= \max(9, 1+8, 5+5, 8+1) \\ &= 10 \end{aligned}$$

- Rod length = 5;

$$\begin{aligned} \text{MaxValue}(5) &= \max(10, \text{MaxValue}(1) + \text{MaxValue}(4), \text{MaxValue}(2) + \\ &\quad \text{MaxValue}(3), \text{MaxValue}(3) + \text{MaxValue}(2), \text{MaxValue}(4) + \\ &\quad \text{MaxValue}(1)) \\ &= \max(10, 1+10, 5+8, 8+5, \\ &\quad 10+1) \\ &= 13 \end{aligned}$$

- Rod length = 6;

$$\begin{aligned} \text{MaxValue}(6) &= \max(17, \text{MaxValue}(1) + \text{MaxValue}(5), \text{MaxValue}(2) + \\ &\quad \text{MaxValue}(4), \text{MaxValue}(3) + \text{MaxValue}(3), \text{MaxValue}(4) + \\ &\quad \text{MaxValue}(2), \text{MaxValue}(5) + \text{MaxValue}(1)) \\ &= \max(17, 1+13, 5+10, \\ &\quad 8+8, 10+5, 13+1) \\ &= 17 \end{aligned}$$

The best price can be achieved by not cutting the rod at all it will be of the highest value that is 17.

TASK 3

```
int Rod-cutting (int price [], int n) {  
    int maxVal[n+1];  
    int cut[n+1];  
  
    maxVal[0] = 0;  
  
    for (int i = 1; i <= n; i++) {  
        int maxVal = -1;  
        for (int j = 1; j <= i; j++) {  
            if (price[j-1] + maxVal[i-j] > maxVal)  
                maxVal = price[j-1] + maxVal[i-j];  
            cut[i] = j;  
        }  
        maxVal[i] = maxVal;  
    }  
  
    // Trace back to find the cut positions  
    int remainingLength = n;  
    while (remainingLength > 0) {  
        cout << cut[remainingLength] << " ";  
        remainingLength -= cut[remainingLength];  
    }  
    cout << endl;  
    return maxVal[n];  
}
```